AIRLINE CUSTOMER EXPERIENCE

Director brief | Lab 11 | 24I-2513 | BDS-5B

Decision: retain human review; use automation for routing

Across 14,485 distinct tweet IDs, VADER agrees with human sentiment on **54.3%**, below the **62.7%** all-negative baseline. It detects only **49.7%** of human-negative tweets. Neutral is hardest by F1 (0.405). A positive score must not close a complaint automatically.

Where customers need help

Common negative phrases include customer service, on hold and cancelled flight. Positive distinctive language includes thanks and great. Prioritize contact-channel delays and disrupted journeys, then verify the underlying incident. Word bars provide more precise comparisons than a word cloud.

Compare shares, but verify the source first

American has the highest customer-service share (39.9% of reason-labelled negative tweets). The supplied Delta category has the highest defined operations share (40.6%), but many of its tweets address @**JetBlue**: confirm the airline mapping before acting. Source text also includes altered strings such as "Cancelled Flightled," which can distort phrase counts.

United has the largest negative count (2,633), but its 68.9% negative share is below US Airways (77.7%) and American (71.6%). Use counts to staff the queue and shares to describe this sample; neither is a complaint rate per passenger.

Actions for the customer-experience team

1. Triage harm first. Review stranded, overnight, safety and accessibility cases first; disruption next; use retweets only as a tie-breaker. Human-check keywords and admit urgent cases even when model sentiment is positive.

2. Protect low-visibility passengers. 93.4% of negative tweets have zero recorded retweets. The top 1% of handles contribute 7.9% of tweets: consolidate conversations and track unique authors alongside volume.

3. Validate before scaling. Obtain the unaltered export and airline mapping. Evaluate a current, adjudicated sample, including sarcasm, negation and domain language. High-confidence labels improve agreement to only 58.6%; keep escalation and quality sampling.

Confidence and limits

The sample covers February 17-24, 2015 (UTC), with partial boundary days. Negative volume peaks on February 23 (2,342); negative share peaks on February 22 (72.9%). Collection volume, repeated authors and missing exposure data prevent causal or passenger-wide conclusions. Twitter participation is self-selected. These are historical findings, not current airline rankings.

Evidence: supplied Tweets.csv; executed notebook Parts A-F. Operations = late/cancelled flights and lost/damaged luggage. All metrics deduplicate tweet IDs; original data is retained.